RAG Interview Questions & Answers

1. Basics

• What is RAG? Retrieval-Augmented Generation combines retrieval with an LLM.

• Why RAG? Reduce hallucinations, use latest/private data, no retraining.

• Pipeline: User Query → Embedding → Vector Search → Top-K Docs → Prompt + Context → LLM → Answer.

• Components: Document Loader, Text Splitter, Embeddings, Vector DB, Retriever, LLM, Prompt Template.

2. Document Loader

Loads documents (PDF, TXT, CSV, DOCX, JSON, Web). Loader reads data; Splitter breaks it into chunks.

3. Chunking

Why? LLM context limit.

Types: Character, Recursive, Token, Sentence, Semantic.

Typical chunk size: 500–1000, overlap: 100–200.

4. Embeddings

Numerical vector representation of text for semantic search.

Models: OpenAI, Sentence Transformers, BGE, E5, Cohere, Gemini.

5. Vector Databases

FAISS, Chroma, Pinecone, Weaviate, Milvus, Qdrant.

Similarity metrics: Cosine, Euclidean, Dot Product.

6. Retriever

Similarity Search, MMR, Hybrid Search, BM25, Dense Retrieval.

Top-K returns most relevant chunks.

7. LangChain

Core: Loaders, Splitters, Embeddings, Vector Stores, Retrievers, Chains, Agents, Memory.

Chain = fixed workflow, Agent = dynamic decisions.

8. Advanced RAG

Hybrid Search, Re-ranking, Context Window, Hallucination, Metadata Filtering, Semantic Search, BM25.

9. Coding Questions

Load PDF, Chunk, Generate Embeddings, Store in FAISS/Chroma/Pinecone, Similarity Search, Conversational RAG.

10. Scenario & Project

Build company chatbot, handle large PDFs, improve retrieval, reduce hallucinations, explain your project, justify vector DB choice, evaluation metrics.